

Tom Sloan

github.com/Tom-Sloan · [linkedin.com/in/tom-sloan](https://www.linkedin.com/in/tom-sloan) · tom-sloan.com

Data Scientist with M.A.Sc. in ECE, specializing in time-series prediction and multi-modal sensor fusion for IoT systems. Experience building production ML pipelines for health monitoring, with expertise in Python, PyTorch/TensorFlow, RAG architectures, and AWS.

Education

Master of Applied Science, Electrical and Computer Engineering

Carleton University, Ottawa - February 2026

Graduate coursework: Machine Learning, Advanced ML, Deep Learning, Computer Vision

Thesis: Indoor 3D Modeling Using Consumer Drones and Neural Simultaneous Localization and Mapping (SLAM) for Virtual Reality

Bachelor of Engineering, Computer Systems Engineering (with Distinction)

Carleton University, Ottawa - 2022 | Dean's Honour List

Technical skills

AI/ML: PyTorch, TensorFlow, Keras, Hugging Face, scikit-learn, supervised/unsupervised learning, regression, classification, time-series forecasting, Neural SLAM

LLM/Agents: RAG pipelines, vector search, prompt engineering, tool-use orchestration, multi-agent systems (Claude, OpenClaw, n8n)

IoT/Sensors: Multi-modal sensor fusion (RGB, IMU, LiDAR, thermal), custom PCB design, embedded firmware

Cloud: AWS (6 certifications including ML Engineer), Docker, RabbitMQ, PostgreSQL, MongoDB, Firebase

Languages: Python (pandas, NumPy, Matplotlib), Rust, C/C++, TypeScript, SQL, React, Node.js

Publications

T. Sloan, B. Wallace, R. Goubran, "Drone-Based Indoor Mapping for Augmented Indoor 3D Modeling Using Neural SLAM," *IEEE ICCE*, Dubai, Feb. 2026

T. Sloan, B. Wallace, R. Goubran, "System for Drone-Based Indoor Mapping for Augmented Reality," *IEEE I2MTC*, Chemnitz, Germany, 2025

T. Sloan, B. Wallace, R. Goubran, "Impact of Image Resolution on Controlling Drones Using Remote VR Headset Visualization and a Cloud Architecture," *IEEE SAS*, Newcastle, UK, 2025

Work experience

Machine Learning Researcher - SAM3 - Nov 2025-Present

- Building predictive models for fall detection, UTI risk, and sleep pattern analysis using multi-modal IoT sensors (pressure mats, thermal cameras, contact sensors) for elderly patient monitoring.
- Developing time-series classification pipelines for AGE-WELL-funded envisAGE aging-in-place study tracking hydration, medication adherence, and daily activity.
- Conducted exploratory analysis of multi-sensor time-series data; presented findings to clinical stakeholders to inform study design.
- Deployed n8n automation pipelines for IoT sensor notifications and trigger-based alerts across distributed sensor nodes.
- Engineered multi-step LLM pipelines using Claude API: system prompt design, tool-use orchestration, and iterative refinement workflows (prompts open-sourced at github.com/Tom-Sloan/llm-research-workflows).

Research Assistant - Carleton University - Oct 2023-Nov 2025

- Co-authored 3 IEEE papers on drone-based indoor mapping; presented at I2MTC 2025, SAS 2025, ICCE 2026.
- Trained and evaluated transformer-based neural SLAM model against traditional approaches, achieving 6.61 cm RMSE on Replica benchmark at 21 FPS inference.
- Open-sourced thesis work: github.com/Tom-Sloan/WorldSystem

DevOps Engineer - Magnet Forensics - Apr 2022-Jun 2023

- Maintained CI/CD infrastructure: 2,000+ daily builds across 30+ Jenkins nodes using Python/PowerShell.
- Migrated build infrastructure to AWS using CloudFormation, reducing deployment friction.
- Improved build reliability by reducing flaky test rates through systematic root cause analysis.

Research Assistant - Carleton University - May 2018-Dec 2021

- Prototyped initial IoT sensor array (pressure mats, thermal cameras, contact sensors) for patient monitoring research.
- Built multi-sensor data collection pipelines for fusion research.

Spectrum Engineering Intern - Telesat - May 2020-Dec 2020

- Developed Python wrapper for satellite spectrum analysis, replacing expensive MATLAB toolboxes.
- Automated ITU regulation compliance testing for spectrum allocation.

Independent projects

Multi-Agent AI System - OpenClaw, Claude, Raspberry Pi - Built dual-agent RAG system: SQLite vector DB with OpenAI embeddings indexing Obsidian vault and Google Drive - Semantic retrieval for context injection; agents coordinate via Telegram for task handoff

Discite - Electron, React, Python, SQL - Automated GDPR data exports from Apple, Google, Discord, Twitter - Built ETL pipelines parsing heterogeneous zip archives; SQL queries for data analysis; visualization dashboards

Smart Home System - C/C++, AWS, Firebase, Supabase, Custom PCBs - Full-stack IoT with custom sensor PCBs, firmware, AWS backend - SQL queries via Supabase for real-time sensor data storage and user account management

Projects open-sourced at tom-sloan.com/#projects

Certifications

- AWS Certified Machine Learning Engineer (Associate) - 2025
- AWS Certified Solutions Architect (Associate) - 2022
- AWS Certified Developer (Associate) - 2022
- *Plus 3 additional AWS certifications (Security Specialty, SysOps, Cloud Practitioner)*